

Class-2

Assignment Questions:

1. Find the finite Fourier Cosine transform of $f(t) = t$.

Ans: $C_0 = \frac{\pi^2}{2}$ and $C_n = \frac{\cos n\pi - 1}{n^2}$.

2. Find the finite Fourier sine transform of $f(t) = \sin(at)$, $a > 0$, if the function is defined on $[0, \pi]$. Also find the inverse finite Fourier Sine transformation.

Ans: $S_n = \frac{1}{2} \left[\frac{\sin(n-a)\pi}{n-a} - \frac{\sin(n+a)\pi}{n+a} \right]$, and

Inverse finite Fourier sine transform is $f(t) = \frac{1}{\pi} \sum_{n=1}^{\infty} \left[\frac{\sin(n-a)\pi}{n-a} - \frac{\sin(n+a)\pi}{n+a} \right] \sin(nt)$.

3. Find the Fourier Cosine Transform of $f(t) = \begin{cases} t, & 0 < t < 1 \\ 2-t & 1 < t < 2. \end{cases}$

Ans: $C_0 = 1$ and $C_n = \frac{1}{n^2 \pi^2} [4 \cos(\frac{n\pi}{2}) - 3 \cos n\pi - 1] - \frac{2}{n\pi} \sin(\frac{n\pi}{2})$.

4. Find the finite Fourier Cosine transform of $f(t) = t^2$, $0 < t < \pi$.

Ans: $C_0 = \frac{\pi^3}{3}$ and $C_n = \frac{2\pi \cos n\pi}{n^2}$.

5. Find the finite Fourier sine transform of $f(t) = \begin{cases} 0, & 0 \leq t < \pi/2 \\ 1, & \pi/2 \leq t \leq \pi \end{cases}$.

Ans: $S_n = \frac{-1}{n} \left[(-1)^n - \cos\left(\frac{n\pi}{2}\right) \right]$.

6. Find the finite Fourier sine transform of $f(t) = 1 - \frac{t}{\pi}$, $0 < t < \pi$.

Ans: $S_n = \frac{1}{n}$.

7. Find the finite Fourier Cosine transform of $f(t) = \frac{\cos k(\pi-t)}{k \sin k\pi}$, $0 < t < \pi$.

Ans: $C_n = \frac{1}{n^2 - k^2}$.